

Consumer Preferences in the Used Vehicle Market: Willingness to Pay for Powertrain Type, Driving Range, Vehicle Condition, and Operating Cost

Zain Hoda^{a,1}, Xiatian Iogansen^{a,b,2}, Christina Gore^{b,*,3}, Joshua D. Kneifel^{b,4},
Sindhu Ranganath^{a,b,5} and John Helveston^{a,6}

^aDepartment of Engineering Management and Systems Engineering, George Washington University, Washington D.C., USA

^bApplied Economics Office, Engineering Laboratory, National Institute of Standards and Technology, USA

ARTICLE INFO

Keywords:

Battery electric vehicle
Used vehicle market
Discrete choice experiment
Willingness to pay
Mixed logit
Consumer preferences

ABSTRACT

This study employs a discrete choice experiment (DCE) to measure consumer willingness to pay (WTP) for powertrain type and vehicle attributes in the used vehicle market in the United States. A nationally distributed online survey was administered to 1,500 respondents who expressed an intention to purchase a used car or sport utility vehicle (SUV) within two years. The DCE presented respondents with repeated choice tasks in which powertrain type (conventional internal combustion engine, gas hybrid electric, or battery electric), purchase price, operating cost, driving range, vehicle age, and mileage were systematically varied. Six mixed logit (MXL) models were estimated, stratified by vehicle segment (Car, and SUV) and budget type (Low, High). Results reveal substantial heterogeneity in BEV preferences within and across segments. Among likely BEV adopters, mean BEV utility is negative but accompanied by large estimated standard deviations, indicating genuine within-group polarization rather than uniform receptivity. Among unlikely adopters, BEV aversion is near-universal, with mean coefficients approximately 15 times larger in magnitude than those of likely adopters, and price sensitivity is 29–63% higher depending on the vehicle segment. Driving range is the largest positive attribute governing BEV utility, particularly among car-segment likely adopters. Vehicle age exerts a large, negative, and monotonic effect on WTP across all segments. The only consumer segment for which estimated WTP reaches or exceeds the price of an equivalent ICE vehicle is likely BEV adopters purchasing 300-mile-range cars at low vehicle age. These findings carry direct implications for used BEV pricing strategy, resale market development, and the design of consumer incentive programs.

1. Introduction

Used vehicles constitute approximately 70% of all vehicle transactions in the United States (Bureau of Transportation Statistics, 2023), yet the used-vehicle segment has received comparatively limited systematic attention from researchers or policymakers. The used-vehicle segment represents a distinct buyer population with different cost constraints, vehicle quality expectations, and information environments, warranting separate empirical attention.

Research on electric vehicle (EV) consumer preferences has concentrated predominantly on new vehicle buyers. The emergence of BEVs in the secondary market adds complexity worth studying. The expanding used BEV market carries significance beyond its transaction volume: it extends vehicle and battery lifetimes, supports a circular economy (Guzek et al., 2024), and lowers the effective cost of EV ownership for lower- and middle-income households (Hagman et al., 2016; Sheykhfar et al., 2025). A healthy secondary market also underpins new-vehicle residual values, which affect the economics of new EV leasing (Brückmann et al., 2021; Lim et al., 2015).

Consumer demand for used BEVs remains constrained by barriers that extend beyond those of the broader EV market. Used BEVs inherit range anxiety, charging access concerns, and upfront cost premium common to all EVs

*Corresponding author

¹zain.hoda@gwu.edu, <https://orcid.org/0009-0003-0958-1868>

²xiatian.iogansen@gwu.edu & xiatian.iogansen@nist.gov, <https://orcid.org/0000-0002-4851-1323>

³christina.gore@nist.gov, <https://orcid.org/0000-0002-3586-6918>

⁴joshua.kneifel@nist.gov

⁵sindhu.ranganath@nist.gov, <https://orcid.org/0000-0001-5764-9773>

⁶jph@gwu.edu, <https://orcid.org/0000-0002-2657-9191>

(Sheykhfard et al., 2025), but face additional friction in the secondary market: uncertainty about prior vehicle condition and reliability is a primary purchase deterrent (Fernandes, 2023), and battery aging introduces uncertainty that age and odometer readings alone cannot fully signal (Prieto et al., 2015).

Understanding the determinants of used BEV purchase decisions is therefore critical for policymakers, manufacturers, dealers, and researchers seeking to support the transition to electric mobility through the secondary vehicle market. This study provides quantitative estimates of willingness to pay (WTP) for powertrain type in the used BEV market in the United States, disaggregated by consumer budget, adoption propensity, vehicle type, driving range, and vehicle age. A discrete choice experiment (DCE) presents respondents with repeated choices between hypothetical used vehicles varying in powertrain, purchase price, operating cost, driving range, vehicle age, and mileage, enabling recovery of implicit monetary valuations for each attribute. To the authors' knowledge, this represents the first study to estimate disaggregated WTP for powertrain type in the U.S. used BEV market across these dimensions simultaneously.

Three dimensions structure the analysis, each grounded in the heterogeneity of the used-vehicle market. Car and SUV buyers operate in different price ranges, have different range requirements, and face different operating cost profiles; pooling them would mask preference structures likely to diverge on the attributes that matter most for BEV adoption. Budget tier determines which vehicles fall within a buyer's realistic consideration set and modulates price sensitivity: lower-budget buyers in the used market face fewer BEV options and bear higher relative cost burdens (Konstantinou et al., 2025; Loh and Noland, 2024), making the price-range interaction a first-order concern. Self-reported BEV adoption propensity captures prior attitude formation and proximity to the purchase threshold, distinguishing consumers for whom attribute improvements may tip the decision from those whose resistance is attitudinal rather than attribute-driven (Moras et al., 2025).

This study addresses two interrelated research questions:

1. How do consumer willingness-to-pay estimates for used-vehicle attributes (powertrain type, driving range, vehicle age, mileage, and operating cost) compare across vehicle segment and budget tier?
2. How do existing used BEVs compete against their conventional or hybrid counterparts as vehicles age?

The remainder of the paper is structured as follows. Section 2 reviews the relevant literature on used-vehicle markets, general EV adoption determinants, and consumer preferences in the used EV segment specifically. Section 3 describes the survey design, data collection, and sample profile. Section 4 presents the statistical methods employed. Section 5 reports and discusses the empirical findings. Section 6 concludes.

2. Literature Review

The review that follows draws on three overlapping bodies of work: the broader used-vehicle market, general EV adoption determinants, and the emerging literature specifically on used EV consumer preferences.

2.1. The Broader Used-Vehicle Market

Affordability, vehicle condition, and reliability are the dominant purchase criteria for used-vehicle buyers (Consumer Reports, 2023; U.S. Environmental Protection Agency, 2023). Fuel costs also factor into vehicle choice, particularly among cost-sensitive households. Used-car purchasing is more common among larger households and renters; low-income single-vehicle renters carry the highest vehicle-cost burdens relative to income (Konstantinou et al., 2025).

Those burdens have grown since 2020. Average used-vehicle prices climbed relative to pre-pandemic levels and stayed elevated through 2024–2025 (Cox Automotive, 2024); in April 2025, the average transaction price was \$25,547 (Kelley Blue Book, 2025). Cox Automotive (2025) find that 74% of used-vehicle buyers evaluated both new and used options, and 71% entered the market without a fixed vehicle preference – a sign that buyers are searching wider under price pressure. C-4 Analytics (2025) report that 43% of consumers would switch brands to get a lower price.

2.2. General Factors in EV Adoption

Purchase price, operating costs, driving range, recharging time, charging infrastructure, environmental benefits, and government incentives all shape EV adoption decisions (Zou et al., 2020). Range and recharging time show statistically significant effects across multiple stated-preference studies (Liao et al., 2017). A meta-regression of 33 of those studies finds that consumers accept shorter range in exchange for denser charging networks, but only when BEV range is already above roughly 250–300 km (Rosales-Tristancho et al., 2024). Below that floor, additional charging

infrastructure does not compensate for range loss – a threshold that is relevant for used EVs, where buyers often discount perceived range due to battery aging. Battery warranty and depreciation rate independently affect purchase intent as well (Li et al., 2020).

2.3. Used Electric Vehicle Adoption

The U.S. used-EV market passed 400,000 units in 2023, but studies of consumer preferences in that market are sparse (Moras et al., 2025). Most assume price sensitivity is the main driver and treat it as uniform. Moras et al. (2025) find three distinct segments among prospective used-EV buyers: those for whom cost is a binding constraint, those apprehensive about adapting to EV technology, and those drawn by a favorable cost-benefit ratio and lower depreciation risk. The segments share a price orientation but arrive at it differently.

The demographic profile of current used-EV owners looks similar to new-EV owners: disproportionately male, middle-aged, highly educated, and higher-income (2024; 2024). Full-time employment and private off-street parking also predict ownership (2024). The used market has not yet reached substantially different or lower-income buyers.

Before purchase, used-EV buyers cite price, battery performance, and charging access as their top concerns (Loh and Noland, 2024). Lower-income buyers are more sensitive to both price and charging availability than higher-income buyers are (Loh and Noland, 2024), and concern about infrastructure correlates with outright reluctance to buy (2024). Prior EV ownership runs the other way: experienced EV drivers are more comfortable buying pre-owned, and new EV drivers say they would consider a used EV when range and home-charging needs are met (Mashrur and Mohamed, 2025).

Used-car buyers and new-car buyers show broadly similar EV preference structures, but used-car buyers are more sensitive to fast-charging time and the proximity of charging to home (Zou et al., 2020). They are less likely to access lease pathways, larger purchase incentives, or home charging setups that new-EV buyers often take for granted. On price, average used-EV prices run \$9,000–\$37,000 depending on range and vehicle class, against roughly \$14,000 for a comparable used gasoline vehicle (Konstantinou et al., 2025) – a gap that price incentives may need to close.

2.4. Synthesis and Research Gap

Used-car buyers are more cost-conscious than new-car buyers, and current market conditions have widened that difference. Price drives used-EV interest, but buyers in that market are not a uniform group: their motivations differ, their income and infrastructure access differ, and so do the weight they assign to battery performance, range, and charging. Battery warranty and depreciation rate shape decisions beyond sticker price.

Existing studies draw predominantly from convenience samples or current owners, limiting generalizability to prospective buyers. Few examine how preference weights shift across income groups, housing types, or vehicle ownership contexts in the used-vehicle segment. This study addresses that gap with a discrete choice experiment that estimates willingness to pay for powertrain type, driving range, vehicle age, operating cost, and purchase price across a nationally distributed sample of prospective used-vehicle buyers, disaggregated by adoption propensity and vehicle segment.

3. Survey Design, Data Collection, and Sample Profile

3.1. Vehicle Choice Experiment

Eligible respondents of this study were U.S. adults interested in buying a used car (sedan or hatchback) or SUV (including crossovers) within two years. The core instrument of this study is a DCE with six repeated choice tasks in which respondents choose among three used vehicle alternatives with systematically varied attributes or a “None of the above” opt-out option in each choice.

Four separate DCE designs were constructed, one per combination of vehicle body type (car, suv) and budget tier (low, high), so that the price and vehicle attributes presented to each respondent corresponded to a plausible purchase context (see Table 1). The DCE attributes are mileage, powertrain type, electric driving range, operating cost, model year, and purchase price. The DCE included three powertrain types: conventional gas vehicles (ICEV), gas hybrid electric vehicle (HEV), and battery electric vehicle (BEV) with varying driving ranges for BEV vehicles. Prior evidence identifies age and mileage as the primary determinants of used vehicle value (Prieto et al., 2015), so both are included in the design, with model year serving as the observable proxy for vehicle age.

Operating cost levels were derived from observed fuel economy data. Vehicle energy consumption (in kWh per mile for BEVs and miles per gallon for ICEVs and HEVs) was sourced from EPA ratings for 2024 model-year vehicles,

1 stratified by powertrain and body type. These figures were combined with the 10th–90th percentile range of state-level
2 electricity rates (\$0.11–\$0.33 per kWh) and retail gasoline prices (\$2.66–\$4.77 per gallon) prevailing in early 2025
3 to produce 20 equally-spaced operating cost values per powertrain–body-type combination. Costs were displayed to
4 respondents in cents per mile alongside the miles-per-gallon equivalent of a conventional gasoline vehicle at the same
5 cost, allowing direct comparison across powertrain types. Operating cost was bounded by powertrain type: ICEV
6 profiles were limited to 9–18 cents per mile, BEV profiles to 3–12 cents per mile, and HEV profiles to 6–12 cents
7 per mile for car alternatives and 9–18 cents per mile for SUV alternatives, reflecting plausible fuel cost ranges for
8 each powertrain. Choice sets were drawn at random from the restricted profile space, with alternatives balanced across
9 powertrain types and dominated profiles excluded.

10 Choice profiles were generated using the `cbcTools` R package (Helveston, 2025). Respondents view the definitions
11 of all attribute before DCE tasks(see Table 1). The table also includes detailed attribute levels. Figure 1 provides an
12 example of the choice task format.

13 Before the DCE section in the survey, the respondents were asked to report their budget for the vehicle. In addition,
14 respondents were asked to select one vehicle image (out of nine images) (conditional on the vehicle body type, car or
15 SUV, that they selected) with a style that is most closely aligned with the next vehicle they are considering purchasing.
16 The selected vehicle image was displayed throughout all choice tasks, holding body-style appearance constant across
17 alternatives.

Table 1
BEV choice experiment attributes and levels.

Attribute	Definition	Levels			
		Low-budget Car (≤\$20,000)	High-budget Car (>\$20,000)	Low-budget SUV (≤\$25,000)	High-budget SUV (>\$25,000)
Mileage	The total number of miles a vehicle has traveled since manufacture.	All levels: 20,000–60,000 miles (5000-mile increments).			
Powertrain	The type of powertrain the vehicle runs on.	All levels: (1) <i>Conventional</i> : Gasoline or other liquid-fuel engine, such as diesel or flex-fuel. (2) <i>Gas hybrid electric vehicle (HEV)</i> : Smaller gasoline engine + electric motor + small battery. The gasoline engine recharges the battery to improve fuel efficiency. (3) <i>Battery electric vehicle (BEV)</i> : Electric motor only. Must be plugged into an electrical outlet to be refueled.			
Electric Range	Maximum distance the BEVs can travel on a full charge. Applies to BEVs only.	50–150 miles (50-mile increments)	100–250 miles (50-mile increments)	150–250 miles (50-mile increments)	200–350 miles (50-mile increments)
Operating Cost	Cost in cents per mile driven of fueling the vehicle. The equivalent fuel efficiency in miles per gallon (MPG) of a conventional gasoline vehicle is displayed in parenthesis.	Car: ICEV: 9–18 cents/mile HEV: 6–12 cents/mile BEV: 3–12 cents/mile		SUV: ICEV: 9–18 cents/mile HEV: 9–18 cents/mile BEV: 3–12 cents/mile	
Model Year	The actual year the vehicle was made.	All levels: 2015–2023 (2-year increments).			
Purchase price	The total cost of the vehicle in dollars, including down payment, monthly payments, taxes, and fees.	\$10,000–\$20,000 (\$2,000 increments)	\$20,000–\$40,000 (\$5,000 increments)	\$15,000–\$25,000 (\$5,000 increments)	\$25,000–\$45,000 (\$5,000 increments)

Question 1 of 6

Assume you are able to purchase a vehicle that looks like the one you selected (see photo). Which of the following versions of that vehicle would you be most likely to purchase? Please compare the key features shown for each option before making your selection.



If these were your only options, which would you choose?

Option 1	Option 2	Option 3	Option 4
<u>Purchase price:</u> \$ 40,000	<u>Purchase price:</u> \$ 45,000	<u>Purchase price:</u> \$ 40,000	Even if these were my best options, I would not choose any of these vehicles
<u>Powertrain:</u>  Conventional	<u>Powertrain:</u>  Gas hybrid	<u>Powertrain:</u>  Battery electric 200 miles on a full charge	
<u>Model year:</u> 2019	<u>Model year:</u> 2023	<u>Model year:</u> 2023	
<u>Mileage:</u> 20,000	<u>Mileage:</u> 25,000	<u>Mileage:</u> 25,000	
<u>Operating cost:</u> 15 cents per mile (25.7 MPG equivalent)	<u>Operating cost:</u> 9 cents per mile (37.9 MPG equivalent)	<u>Operating cost:</u> 9 cents per mile (97.2 MPG equivalent)	

Figure 1: Example of a vehicle choice task presented to respondents. Each task offers three used vehicle alternatives with varying attributes and a “None of the above” opt-out option.

3.2. Data Collection and Sample Profile

Data were collected online between December 2025 and April 2026 via two panel providers (Dynata and Prolific), with participants compensated upon completion. Following initial processing, 2,113 records were retained from sessions with a median completion time of approximately 14 minutes. Responses were then screened for attention-check failures, cross-section inconsistencies, and incoherent open-ended text, yielding a final analytic sample of 1,500 respondents.

Table 2 compares the analytic sample against U.S. population benchmarks from the American Community Survey (ACS) 2020–2024 five-year estimates. The sample skews younger, more female, more educated, and more likely to rent than the general adult population, consistent with the online panel recruitment method and the used-vehicle buyer focus. Income distribution approximates the national profile at middle brackets, while the highest income tier (\$150,000 or more) is underrepresented relative to ACS.

Table 2

Sample profile compared with the U.S. population (ACS 2020–2024 5-year estimates).

	Sample (n)	Sample (%)	ACS (%)
<i>Age</i>			
18 - 24	329	14.7	11.7
25 - 34	675	30.2	17.5
35 - 44	574	25.7	16.9
45 - 54	344	15.4	15.6
55 - 64	190	8.5	16.2
65+	121	5.4	22.0
<i>Gender</i>			
Female	1,340	59.6	51.0
Male	861	38.3	49.0
Other	49	2.2	–
<i>Race</i>			
White alone	1,543	68.5	63.3
Black or African American alone	322	14.3	11.9
Other or multiple races	389	17.3	24.8
<i>Hispanic origin</i>			
Hispanic or Latino	269	11.9	17.4
Not Hispanic or Latino	1,985	88.1	82.6
<i>Household income</i>			
Less than \$25,000	343	15.2	14.6
\$25,000 - \$49,999	374	16.6	16.7
\$50,000 - \$74,999	545	24.2	15.5
\$75,000 - \$99,999	257	11.4	12.6
\$100,000 - \$149,999	428	19.0	17.5
\$150,000 or more	304	13.5	23.0
<i>Education</i>			
High school or less	314	14.0	36.4
Some college or associate	731	32.5	27.9
Bachelor's degree	765	34.0	21.6
Graduate or professional degree	438	19.5	14.1
<i>Housing tenure</i>			
Own	1,087	52.0	65.2
Rent	1,002	48.0	34.8
<i>Housing type</i>			
Single-family detached	1,367	60.6	62.3
Single-family attached	227	10.1	6.4
Apartment (2+ units)	574	25.5	26.0
Mobile home or other	86	3.8	5.2

Notes: Sample size and shares are unweighted and exclude item nonresponse. '–' indicates no comparable ACS category. ACS universes: population 18+ (age, gender, race, Hispanic origin), population 25+ (education), and households (income, tenure, housing type).

3.3. Household Vehicle Context and Used-BEV Purchase Interest

Before the DCE, respondents reported their current household vehicle holdings and several factors related to EV exposure and purchase interest. Table 3 summarizes these measures. Most respondents' households own at least one vehicle: 35.2% report one vehicle and 37.9% report two, while 4.0% report owning none. Among respondents reporting a current primary household vehicle, the large majority (83.8%) drive a conventional gasoline vehicle, with hybrids (10.2%), battery electric vehicles (4.1%), and plug-in hybrids (1.8%) comprising the remainder – consistent with the sample's used-vehicle-buyer focus rather than existing EV ownership. Home charger access is split roughly evenly, with 44.0% reporting access to a home charger and 49.1% reporting none. A comparable share of respondents report knowing a neighbor who owns an EV (39.8%) as report not knowing one (31.9%), with 28.3% unsure. Stated likelihood

1 of purchasing a used BEV as their next vehicle is concentrated at the unlikely end of the scale: 34.3% report being very
2 unlikely and 20.8% somewhat unlikely to do so, versus 8.5% very likely and 22.8% somewhat likely, with the remaining
3 13.6% neutral. These pre-DCE measures motivate the adoption-propensity subgroup analysis in Sections 5.1 and 6.1.

Table 3

Household vehicle context and used-BEV purchase interest.

	n	%
<i>Household vehicle count</i>		
0	91	4.0
1	793	35.2
2	854	37.9
3	292	13.0
4 or more	223	9.9
<i>Current primary vehicle fuel type</i>		
Gasoline (ICEV)	1,153	83.8
Hybrid (HEV)	140	10.2
Plug-in hybrid (PHEV)	25	1.8
Battery electric (BEV)	57	4.1
Other	1	0.1
<i>Home charger access</i>		
Yes	992	44.0
No	1,106	49.1
Not sure	154	6.8
<i>Knows a neighbor who owns an EV</i>		
Yes	896	39.8
No	718	31.9
Not sure	638	28.3
<i>Likelihood of purchasing a used BEV next</i>		
Very unlikely	773	34.3
Somewhat unlikely	469	20.8
Neutral	307	13.6
Somewhat likely	514	22.8
Very likely	191	8.5

Notes: n = 2,254 unless otherwise noted; percentages exclude item nonresponse. Current primary vehicle fuel type was asked only of respondents reporting a current household vehicle (n = 1,376); all other items were asked of the full analytic sample.

4. Method

4.1. Mixed Logit Models

Consumer choice is modeled within the random utility framework. Each respondent i is assumed to select the alternative j that maximizes their latent utility U_{ij} in each choice task, where utility is a linear function of observed vehicle attributes augmented by an unobserved idiosyncratic error term assumed to follow an independent Gumbel distribution.

A multinomial logit (MNL) model provides the baseline specification, but two features of the choice setting favor the mixed logit (MXL) extension. First, the MNL imposes the Independence of Irrelevant Alternatives (IIA) property, which requires consumers to redistribute among remaining options in fixed proportion to current shares whenever one alternative is removed. With qualitatively distinct powertrains, this is implausible: a consumer who prefers a conventional ICEV would, if that option were unavailable, substitute far more toward an HEV than a BEV, because the two share a combustion engine and a familiar fueling experience. Second, the MNL assumes preference homogeneity across respondents — a restriction inconsistent with the substantial heterogeneity in EV attitudes documented in the literature, and rejected empirically by statistically significant standard deviation parameters $\hat{\sigma}_k$ across virtually all attributes in all models. The MXL addresses both problems by modeling each preference parameter as an independent draw from a normal mixing distribution:

$$\beta_{ik} \sim \mathcal{N}(\mu_k, \sigma_k^2) \quad (1)$$

where μ_k is the population mean valuation for attribute k and σ_k captures the dispersion of that valuation across respondents. A statistically significant $\hat{\sigma}_k$ constitutes evidence that preferences for attribute k are genuinely heterogeneous in the population rather than uniform. The seven parameters specified as randomly distributed are the BEV powertrain indicator, HEV powertrain indicator, BEV driving range, vehicle mileage, vehicle age, operating cost, and the alternative-specific constant for the no-choice (opt-out) option. Purchase price enters as the scale parameter λ_i in the WTP-space reparameterization rather than as a randomly distributed preference weight. The internal combustion engine vehicle (ICEV) serves as the reference powertrain category throughout.

Rather than deriving WTP post-estimation as a ratio of coefficients, the models are specified directly in WTP space by reparameterizing the utility function as:

$$U_{ij} = \lambda_i \left(\sum_k \omega_{ik} x_{ijk} - p_{ij} \right) + \varepsilon_{ij} \quad (2)$$

Six MXL models are estimated across two stratification dimensions: two primary vehicle-segment models (car and SUV), four budget-tier subgroup models (car and SUV each crossed with *low* and *high* budget). All models are estimated in R using the `logitr` package in willingness-to-pay space, with a panel correction applied via respondent identifiers to account for the correlation of the six repeated choice observations per respondent. The mixing integral is approximated using 5,000 Sobol quasi-random draws per respondent. To reduce sensitivity to starting values and mitigate convergence to local optima, each model is initialized from 10 independent random starting points; the run achieving the globally lowest log-likelihood is retained. Convergence is verified via the relative and absolute function tolerance criterion (Exit Status 3) in all models.

where $\lambda_i > 0$ is a respondent-specific price sensitivity (scale) parameter and ω_{ik} is respondent i 's marginal WTP for attribute k , expressed directly in dollar units. Each ω_{ik} is drawn from a normal mixing distribution $\mathcal{N}(\mu_{\omega_k}, \sigma_{\omega_k}^2)$, so the population mean WTP for attribute k is recovered directly from $\hat{\mu}_{\omega_k}$ without ratio computation. Uncertainty bounds are obtained by drawing 10,000 samples from the estimated joint parameter distribution and reporting the 95% simulation interval.

4.2. WTP for Defined Vehicle Profiles

While WTP-space estimation yields interpretable marginal valuations for each attribute, reporting coefficients in isolation is abstract for powertrain-type comparisons. A consumer choosing between a used BEV and an equivalent ICEV does not weigh powertrain type and driving range as separate attributes: range is a property of the BEV that does not apply to the ICEV baseline. Reporting the BEV powertrain coefficient alone corresponds to WTP for a BEV with zero driving range, which is economically meaningless. To produce interpretable figures, BEV WTP is therefore computed for specific vehicle configurations by combining the powertrain, range, and age parameters:

$$\text{WTP}_{\text{BEV}}(r, a) = \hat{\mu}_{\omega_{\text{BEV}}} + \hat{\mu}_{\omega_{\text{range}}} \cdot r + \hat{\mu}_{\omega_{\text{age}}} \cdot a \quad (3)$$

where r is driving range in hundreds of miles and a is vehicle age in years. This quantity represents the dollar premium (positive) or required discount (negative) for a consumer to be indifferent between a BEV with range r and age a and an otherwise identical ICEV. Results sections 5.2 through 5.4 present this combined WTP at fixed combinations of range (100, 200, and 300 miles) and vehicle age (2, 5, and 8 years). Uncertainty is propagated by drawing 10,000 samples from the estimated joint parameter distribution and reporting 95% simulation intervals.

5. Results

5.1. Mixed Logit Estimates: Preference Heterogeneity by subgroups

Table 4 reports the full MXL parameter estimates for the six models stratified by vehicle segment (car, SUV) and by budget tier within segment (car low, car high, SUV low, SUV high). For each attribute, the population mean coefficient ($\hat{\mu}$) and the estimated standard deviation of the mixing distribution ($|\hat{\sigma}|$) are presented, both expressed in \$10,000 WTP-space units. A statistically significant $|\hat{\sigma}|$ indicates genuine preference heterogeneity for that attribute within the subgroup. Model fit statistics are reported at the base of the table.

Table 4

Mixed logit model estimates in WTP space: by vehicle type and budget subgroups (price unit: \$10,000).

	By Vehicle Type		By Budget			
	Car	SUV	Car (Low)	Car (High)	SUV (Low)	SUV (High)
Mean parameters						
λ (scale)	1.733 (0.058)***	1.364 (0.045)***	3.649 (0.136)***	1.220 (0.063)***	2.452 (0.117)***	1.202 (0.050)***
BEV powertrain	-1.729 (0.120)***	-2.565 (0.197)***	-1.030 (0.082)***	-1.858 (0.271)***	-2.044 (0.197)***	-2.267 (0.297)***
HEV powertrain	0.067 (0.047)	0.079 (0.055)	-0.061 (0.032)	0.421 (0.101)***	-0.070 (0.050)	0.231 (0.081)**
Electric range	0.434 (0.065)***	0.451 (0.065)***	0.213 (0.056)***	0.542 (0.120)***	0.412 (0.078)***	0.374 (0.094)***
Mileage	-0.203 (0.012)***	-0.302 (0.016)***	-0.100 (0.008)***	-0.314 (0.027)***	-0.148 (0.014)***	-0.381 (0.025)***
Vehicle age	-0.083 (0.007)***	-0.144 (0.009)***	-0.028 (0.004)***	-0.191 (0.017)***	-0.044 (0.007)***	-0.216 (0.015)***
Operating cost	-0.749 (0.051)***	-0.858 (0.059)***	-0.413 (0.033)***	-1.050 (0.114)***	-0.582 (0.053)***	-0.897 (0.085)***
No choice	-5.756 (0.144)***	-7.456 (0.173)***	-3.304 (0.084)***	-7.783 (0.314)***	-4.676 (0.153)***	-8.452 (0.245)***
Standard deviation						
$ \hat{\sigma} $: BEV powertrain	1.684 (0.097)***	2.226 (0.129)***	0.943 (0.064)***	2.214 (0.214)***	1.462 (0.113)***	2.306 (0.215)***
$ \hat{\sigma} $: HEV powertrain	0.982 (0.055)***	1.234 (0.068)***	0.554 (0.034)***	1.171 (0.121)***	0.724 (0.058)***	1.388 (0.100)***
$ \hat{\sigma} $: Electric range	0.454 (0.096)***	0.208 (0.120)	0.226 (0.107)*	0.607 (0.154)***	0.004 (0.180)	0.330 (0.143)*
$ \hat{\sigma} $: Mileage	0.144 (0.017)***	0.205 (0.021)***	0.076 (0.010)***	0.172 (0.041)***	0.114 (0.018)***	0.229 (0.030)***
$ \hat{\sigma} $: Vehicle age	0.097 (0.010)***	0.130 (0.013)***	0.044 (0.007)***	0.168 (0.021)***	0.046 (0.014)**	0.174 (0.018)***
$ \hat{\sigma} $: Operating cost	0.556 (0.062)***	0.746 (0.067)***	0.305 (0.036)***	0.677 (0.147)***	0.422 (0.060)***	0.836 (0.097)***
$ \hat{\sigma} $: No choice	1.389 (0.097)***	1.604 (0.122)***	0.495 (0.070)***	1.725 (0.217)***	0.899 (0.111)***	1.511 (0.173)***
Respondents	1,099	1,155	657	442	441	714
Choice observations	6,594	6,930	3,942	2,652	2,646	4,284
Log-Likelihood	-6,762.75	-7,198.82	-3,759.47	-2,726.68	-2,620.13	-4,448.66
Null Log-Likelihood	-9,141.23	-9,607.02	-5,464.77	-3,676.45	-3,668.13	-5,938.89
AIC	13,555.50	14,427.64	7,548.95	5,483.35	5,270.26	8,927.32
BIC	13,657.41	14,530.30	7,643.14	5,571.60	5,358.48	9,022.76
McFadden R^2	0.2602	0.2507	0.3121	0.2583	0.2857	0.2509
Adj. McFadden R^2	0.2586	0.2491	0.3093	0.2543	0.2816	0.2484

Notes: WTP estimates in \$10,000 units. Significance codes: $p < 0.001$ '***', $p < 0.01$ '**', $p < 0.05$ '*', $p < 0.1$ '.'. Standard errors in parentheses.

The sign and significance pattern of the mean coefficients is remarkably consistent across all six specifications. BEV powertrain utility is negative and highly significant everywhere ($\hat{\mu} = -1.030$ to -2.565 , $p < 0.001$ throughout), more negative for SUV buyers than car buyers at every budget tier, and more negative for high-budget than low-budget buyers within the car segment ($\hat{\mu} = -1.858$ vs. -1.030); the budget gradient is comparatively muted within the SUV segment ($\hat{\mu} = -2.267$ vs. -2.044). HEV powertrains follow a different pattern: statistically indistinguishable from the conventional baseline among low-budget buyers in both segments, but carrying a significant premium among high-budget buyers (car high: $\hat{\mu} = 0.421$, $p < 0.001$; SUV high: $\hat{\mu} = 0.231$, $p < 0.01$) – hybrid acceptance concentrates in the higher-budget market rather than being distributed evenly. Electric driving range is positive and significant in all six models, and mileage, vehicle age, and operating cost are negative and significant in all six, with each attribute's magnitude scaling consistently with both budget tier (larger for high-budget buyers) and vehicle segment (larger for SUV than car buyers). Model fit is comparable across specifications (McFadden $R^2 = 0.249$ – 0.312), with the strongest fit obtained for the low-budget car model. Section 6.1 translates these coefficients, which are already expressed in WTP space, into the dollar-denominated valuations reported in Figure 2.

Beyond the mean effects, the standard deviation estimates establish that these are population-average patterns masking substantial within-subgroup heterogeneity, not uniform preferences – a distinction the mean-only WTP figures in Section 6.1 cannot show. BEV and HEV powertrain coefficients carry large, statistically significant standard deviations in all six models ($|\hat{\sigma}|_{\text{BEV}} = 0.943$ – 2.306 ; $|\hat{\sigma}|_{\text{HEV}} = 0.554$ – 1.388 ; $p < 0.001$ throughout), confirming that the negative mean BEV valuation coexists with a meaningful share of respondents holding substantially more favorable views within every segment and budget tier. Mileage, age, and operating cost heterogeneity is likewise significant in every subgroup. Driving range is the one attribute for which this pattern breaks down: heterogeneity is significant for car buyers overall and both car budget tiers ($|\hat{\sigma}| = 0.226$ – 0.607 , $p < 0.05$), only marginally significant for SUV buyers overall ($|\hat{\sigma}| = 0.208$, $p < 0.1$), and statistically indistinguishable from zero among low-budget SUV buyers ($|\hat{\sigma}| = 0.004$, n.s.) – range preferences are comparatively homogeneous within that one subgroup, in contrast to the significant dispersion found for every other attribute throughout the table. Sections 6.1 and 6.2 build on these mean estimates to quantify what the coefficients imply in dollar terms and for real-vehicle market competitiveness, respectively.

6. Analysis

6.1. Q1 - How do the user preferences for vehicle attributes compare?

Figure 2 summarizes estimated willingness to pay for each vehicle attribute, disaggregated by vehicle segment (car, SUV) and budget tier (low, high budget); error bars represent 95% confidence intervals around the mean WTP estimates, and heterogeneity in preferences across individuals is not shown. For the three BEV configurations (100, 200, and 300 miles of range), solid error bars denote WTP calculated at range levels that fall within the range shown to that segment in the choice experiment, whereas dotted error bars denote WTP extrapolated to range levels outside that segment's design space and should be interpreted with greater caution. The figure axis is scaled in thousands of dollars for readability; dollar figures below are reported at full precision. For powertrain type, high-budget buyers place a statistically significant premium on HEVs relative to the conventional gasoline baseline: approximately \$4,200 among car buyers and \$2,300 among SUV buyers. Low-budget buyers, by contrast, show no significant HEV premium in either segment, with the car-segment estimate even weakly negative (approximately -\$600, $p < 0.1$). BEV powertrains carry a WTP penalty relative to conventional vehicles in every segment and budget combination, though the size of the penalty narrows sharply as driving range increases. At 100 miles of range, WTP for a BEV is approximately -\$13,200 among high-budget car buyers, -\$8,200 among low-budget car buyers, -\$18,900 among high-budget SUV buyers, and -\$16,300 among low-budget SUV buyers. Extending range to 300 miles closes most of this gap: WTP rises to approximately -\$2,300 (high-budget car, with a confidence interval spanning zero), -\$3,900 (low-budget car), -\$11,500 (high-budget SUV), and -\$8,100 (low-budget SUV). The 300-mile car result for high-budget buyers is the only combination in the figure for which BEV WTP is not statistically distinguishable from the conventional baseline, identifying added range as the single most effective lever for closing the BEV valuation gap, particularly for higher-budget car buyers.

Vehicle age, mileage, and operating cost are each valued negatively across all four subgroups, and each penalty scales consistently with budget tier: high-budget buyers exhibit larger dollar-denominated penalties than low-budget buyers for the same attribute in both segments. Operating cost carries the largest of the three penalties throughout, ranging from approximately -\$4,100 (low-budget car) to -\$10,500 (high-budget car), and from -\$5,800 (low-budget SUV) to -\$9,000 (high-budget SUV). Vehicle age WTP ranges from approximately -\$300 (low-budget car) to -\$1,900

(high-budget car), and from -\$400 (low-budget SUV) to -\$2,200 (high-budget SUV); mileage follows a comparable three- to four-fold gradient between budget tiers within each segment. This pattern – larger dollar penalties among higher-budget buyers – mirrors the overall scale of WTP in each subgroup rather than reflecting a qualitatively different preference structure.

Comparing across vehicle segments, SUV buyers exhibit systematically larger BEV penalties and larger operating-cost, mileage, and age penalties than car buyers at the same budget tier, while HEV premiums are smaller for SUV buyers than for car buyers among high-budget respondents. Together these patterns indicate that BEV valuation gaps are narrowest, and closest to parity with conventional vehicles, among higher-budget car buyers purchasing longer-range vehicles – consistent with the preference structure reported in Table 4.

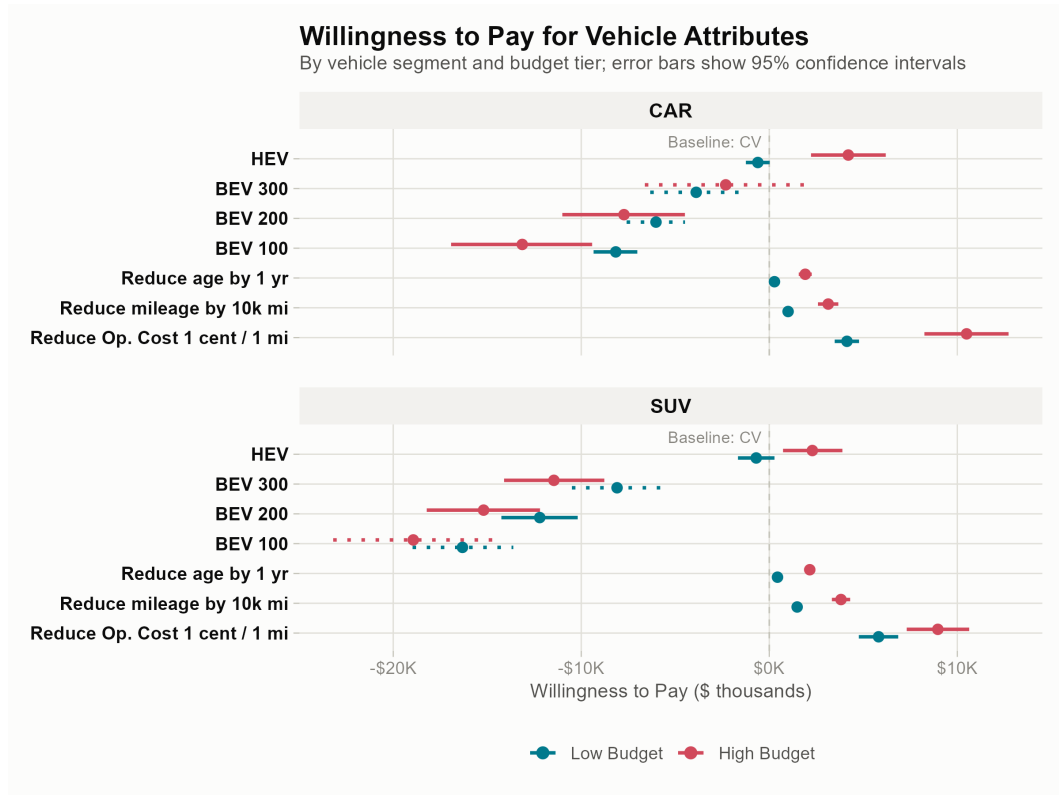


Figure 2: Willingness to pay for vehicle attributes, by vehicle segment and budget tier (95% confidence intervals shown; values in thousands of dollars). For the BEV range configurations, solid intervals mark WTP calculated within the range shown to that segment and dotted intervals mark WTP extrapolated beyond it.

6.2. Q2 - How do the existing used BEVs compete against their conventional counterparts?

Figure 4 extends the willingness-to-pay estimates in Section 6.1 to a direct market-share simulation, following the approach of Helveston et al. (2014): rather than comparing isolated attribute coefficients, the estimated MXL models are used to predict a two-alternative choice probability between a real BEV and its closest currently marketed conventional or hybrid counterpart. Five matched pairs of real, currently marketed vehicles were selected for comparable body style and market positioning – three in the car segment and two in the SUV segment, spanning both BEV-vs-CV and BEV-vs-HEV comparisons. For each vehicle, real used-market listings (model years 2017–2024, used inventory only) were used to fit a hedonic depreciation regression of log price on mileage and vehicle age; the fitted regression generates a predicted market price at each vehicle age from 0 to 8 years, with mileage held at 10,000 miles per year of age. BEV driving range is depreciated at an assumed 5% annual rate, and operating cost is fixed at the powertrain-typical value used elsewhere in the design (3, 6, and 12 cents per mile for BEV, HEV/PHEV, and CV alternatives, respectively). These age-specific attribute profiles are passed to the vehicle-segment- and budget-tier-matched MXL model from Table 4, using the low- or high-budget model corresponding to each pair's DCE design tier ($\leq \$20,000$ or $> \$20,000$),

to predict the probability that a respondent chooses the BEV over its real-world counterpart at each vehicle age. Rather than naming specific makes and models, Figure 4 labels each pair generically by vehicle segment, the budget tier of the matched DCE design, and the powertrain comparison it represents (BEV vs. CV or BEV vs. HEV). Uncertainty in the predicted probabilities is propagated by drawing 10,000 samples from each model's estimated joint parameter distribution; the shaded band around each curve is the resulting 95% simulation interval.

One of the three car-segment pairings is competitive with, or ahead of, its conventional counterpart even when new: the high-budget car pairing (BEV vs. CV) already holds a narrow majority (49% predicted share) at age 0 and rises to 57% by age 2 (its real-market listings span only three model years, so its curve does not extend further). The remaining two car pairings start below parity but close much of the gap as vehicles age: the mid-budget car pairing (BEV vs. HEV) rises from 22% to 34% share between ages 0 and 5, and the low-budget car pairing (BEV vs. CV) shows the steepest gain of all five pairs, from 6% share at age 0 to 36% by age 7. Neither SUV pairing reaches parity within the observed age range, though both trend upward: the mid-budget SUV pairing (BEV vs. CV) rises from 15% to 35% share between ages 0 and 5, overtaking the high-budget SUV pairing (BEV vs. HEV, which rises more modestly, from 24% to 32%) by around four years of age.

The rising BEV share with vehicle age in every one of the five pairings runs counter to the age effect reported in Table 4, where vehicle age depresses WTP for BEV and conventional vehicles alike. The reconciliation is that the two vehicles in each pair do not depreciate at the same real-market rate: the hedonic regressions underlying Figure 4 show BEV listing prices falling faster with age than their conventional or hybrid counterparts', so the widening price discount on the aging BEV increasingly offsets both the fixed BEV powertrain penalty and the shared age penalty estimated in the MXL models. This is consistent with the logic in Helveston et al. (2014): competitiveness is set by the full attribute bundle a specific vehicle offers relative to its counterpart, not by powertrain type alone, and in the used market, faster BEV price depreciation is itself an attribute that increasingly favors the BEV as both vehicles age.

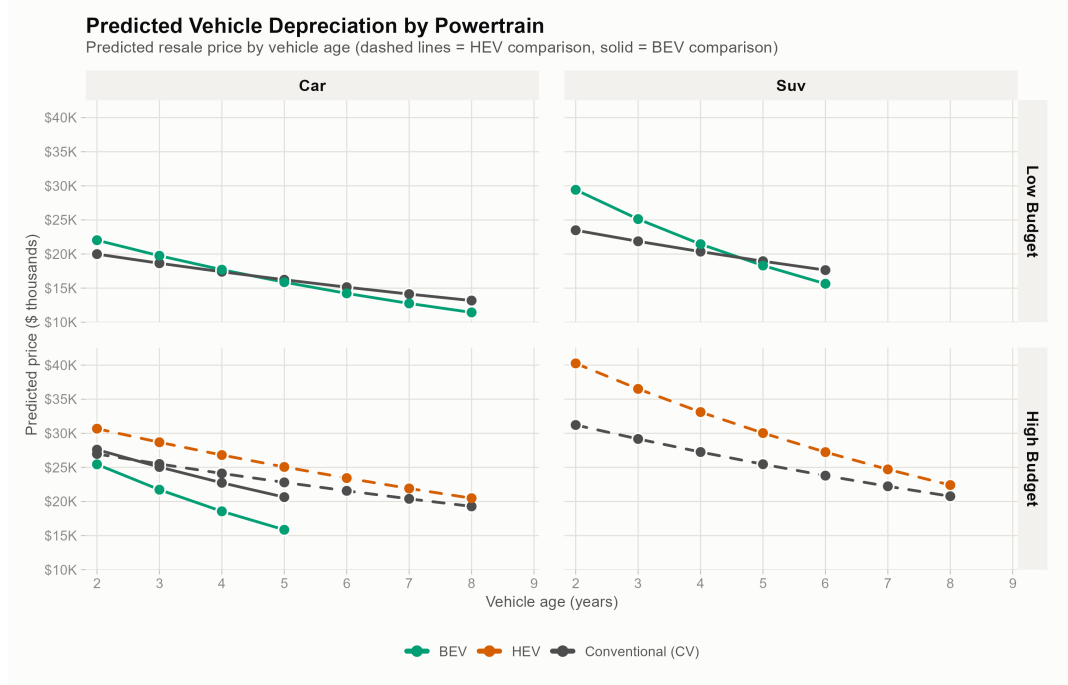


Figure 3: Predicted used-market resale price by vehicle age and powertrain for the matched vehicle pairs, from hedonic regressions of log price on mileage and vehicle age (solid lines: BEV-versus-conventional pairs; dashed lines: HEV-versus-conventional pairs)

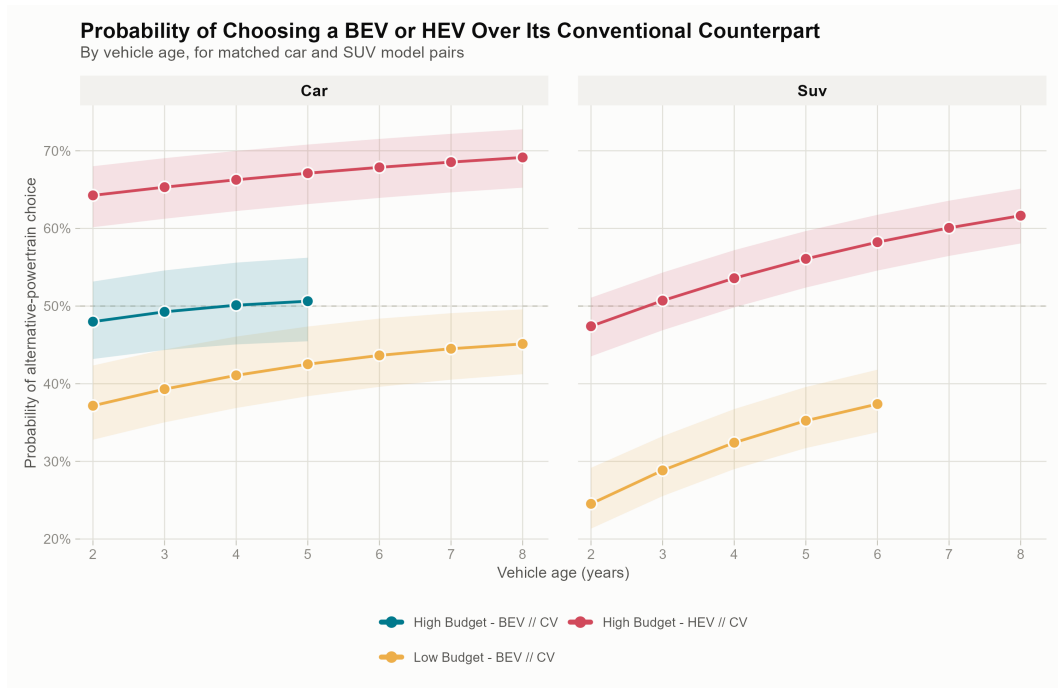


Figure 4: Probability of choosing a used BEV or HEV over its matched conventional counterpart, by vehicle age, vehicle segment, and budget tier. Lines are mean predicted two-alternative choice probabilities and shaded bands are 95% simulation intervals from 10,000 draws of the estimated parameter distribution; the dashed horizontal line marks choice parity (50%).

7. Discussion: Policy Interventions in Light of the Estimates

Most U.S. incentives for electric vehicle adoption were designed around the new-vehicle market. The federal clean vehicle credit (Internal Revenue Code §30D) has long offered up to \$7,500 toward a qualifying new EV, and a used-vehicle counterpart arrived only with the Inflation Reduction Act of 2022, which established the used clean vehicle credit (§25E): the lesser of \$4,000 or 30% of the sale price, limited to vehicles priced at or below \$25,000 and to buyers below set income thresholds, and transferable to the dealer as a point-of-sale discount since 2024. State programs, from California’s Clean Vehicle Rebate Project to scrappage-and-replacement schemes such as Clean Cars 4 All, follow a similar structure. Read against the estimates in Section 6.1, the \$4,000 ceiling matters for only a narrow part of the market. It is comparable to the residual valuation gap for high-budget car buyers considering longer-range vehicles at low age, the segment already closest to parity, but is far below what other segments require: roughly \$8,000–\$13,000 to close the gap on a 100-mile used BEV in the car segment and \$16,000–\$19,000 in the SUV segment, and \$4,000–\$12,000 even for a 300-mile BEV. Unlikely BEV adopters, whose aversion is near-universal and whose price sensitivity is 29–63% higher than that of likely adopters, require discounts several times larger still, so a fixed \$4,000 transfer does little to change their choice.

The design of the credit also disfavors the most responsive segment. The \$25,000 price cap excludes most low-age used BEVs with 250–300 miles of range, the configuration for which WTP reaches conventional-vehicle parity among high-budget car buyers, while directing support toward shorter-range or older vehicles whose valuation gap \$4,000 cannot close; applied instead to the near-parity segment, an unconditioned subsidy would mostly reward purchases that would have occurred anyway. Consistent with Roberson and Helveston (2022), incentive structure matters as much as magnitude: buyers value immediate rebates well above equivalent tax credits, and that premium is largest for the lower-income, lower-budget, and used-vehicle buyers who dominate this market. The estimates also point to non-price instruments. About \$15,000–\$20,000 of the WTP difference between a two- and an eight-year-old BEV exceeds conventional depreciation and most plausibly reflects uncertainty about battery state of health; standardized health disclosure, third-party certification, or transferable extended warranties (Li et al., 2020) would address that mechanism directly and at lower fiscal cost than a cash transfer. Because driving range raises utility for likely adopters but not for the most resistant car buyers, and because a receptive HEV minority is evident among unlikely SUV adopters, the

results favor segment-aware policy: point-of-sale incentives targeted by vehicle age and range, paired with measures that close the battery-health information gap.

8. Conclusion

Driving range, vehicle age, and adoption preference are the dominant value determinants, each producing WTP effects that substantially exceed those of budget tier and vehicle type. The only segment for which WTP reaches or exceeds the ICE-parity threshold is likely BEV adopters selecting 300-mile-range cars at two years of age. For all other attribute combinations, used BEVs must be offered at a discount, with the required discount scaling predictably with increasing vehicle age, declining range, and greater adoption reluctance.

This study provides quantitative estimates of consumer willingness to pay for used battery electric vehicles in the United States, disaggregated by adoption propensity, vehicle type, driving range, vehicle age, and budget tier. A discrete choice experiment administered to 1,500 prospective used-vehicle buyers yielded six mixed logit models that recover preference parameters and characterize heterogeneity across adoption and vehicle segments.

Three findings carry direct implications for used BEV market development. First, the used BEV market has not reached price parity with conventional vehicles for the majority of prospective buyers: only likely BEV adopters selecting 300-mile-range cars at two years of age demonstrate WTP at or near the ICE-equivalent price, while all other segment combinations require discounts ranging from modest to over \$35,000. Second, vehicle age is a first-order negative driver of WTP across all segments, with an estimated WTP differential of \$15,000 to \$20,000 between 2-year-old and 8-year-old vehicles within a given segment. This excess discount relative to standard ICE depreciation most plausibly reflects consumer uncertainty about battery state-of-health, suggesting that interventions improving SOH transparency (standardized health disclosures, third-party certification, or extended manufacturer warranties in the secondary market) could recover a portion of this penalty without requiring changes to the vehicle itself. Third, driving range is the primary positive attribute for likely BEV adopters, particularly in the car segment, yet range has no statistically significant effect among car-segment unlikely adopters, establishing that range improvements alone are insufficient to expand adoption into the most resistant consumer group.

These findings point to a segmented market in which no single pricing or policy instrument is uniformly effective. For likely adopters (the population proximate to the purchase threshold), communication strategies emphasizing driving range sufficiency and per-mile operating cost advantages are the most efficient levers for conversion, given this group's high purchase engagement and elevated operating cost sensitivity. For unlikely adopters, whose BEV aversion is near-universal and price sensitivity is 29–63% higher than among likely adopters depending on the segment, the federal used EV tax credit (currently capped at \$4,000) falls well short of the discounts required to reach purchase indifference. The most tractable near-term intervention for this population is the promotion of hybrid electric vehicles, particularly in the SUV segment, where within-group HEV heterogeneity indicates a non-trivial receptive minority. State-level programs designed with segment awareness, concentrating on high-budget car buyers at low vehicle age where the market is already proximate to parity, would achieve better marginal additionality than uniform subsidies applied irrespective of vehicle age, range, or buyer propensity.

Several limitations bound the scope of these findings. The analysis rests on stated rather than revealed preferences; hypothetical bias may affect WTP magnitudes, though its direction in this context is uncertain. The sample was drawn from online research panels and may overrepresent information-engaged consumers. The MXL specification assumes continuous normal mixing distributions, which may inadequately represent discrete preference clusters; a latent class extension would test this directly. Battery state-of-health was not varied in the vehicle powertrain DCE analyzed here and is addressed in a companion battery DCE component of the broader study. Integrating findings from both instruments and cross-validating against revealed transaction prices from dealer records would substantially strengthen the empirical basis for the policy inferences drawn above.

LIMITATIONS: - PHEV is not included in DCE, why?

- FUTURE WORK: - Many information collected in the survey was not utilized in this study. The study can further segment individuals based on other variables.

CRedit authorship contribution statement

Zain Hoda: Conceptualization, Data curation, Methodology, Software, Formal analysis, Writing - Original Draft, Writing - Review & Editing. **Xiatian Iogansen:** Conceptualization, Data curation, Methodology, Writing - Review

& Editing. **Christina Gore:** Conceptualization, Data curation, Methodology, Investigation, Writing - Original draft preparation, Writing - Review & Editing, Supervision, Project administration, Funding acquisition. **Joshua D. Kneifel:** Conceptualization, Data curation, Methodology, Investigation, Writing - Review & Editing, Supervision, Project administration, Funding acquisition. **Sindhu Ranganath:** Data curation, Writing - Original Draft, Writing - Review & Editing. **John Helveston:** Conceptualization, Methodology, Writing - Review & Editing, Supervision.

References

- (2024). In: *Transportation Research Part D: Transport and Environment*. Used-EV owner demographics: disproportionately male, middle-aged, well-educated, higher-income. URL: <https://www.sciencedirect.com/science/article/pii/S1361920924000439>.
- (2024). In: *Transportation Research Part D: Transport and Environment*. High-income, full-time-employed owners with private parking more likely to buy used EVs; charging infrastructure and price concerns as barriers. URL: <https://www.sciencedirect.com/science/article/pii/S1361920924005121>.
- Brückmann, G., M. Wicki, and T. Bernauer (Nov. 2021). "Is Resale Anxiety an Obstacle to Electric Vehicle Adoption? Results from a Survey Experiment in Switzerland". In: *Environmental Research Letters* 16.12, p. 124027. ISSN: 1748-9326. DOI: 10.1088/1748-9326/ac3531. URL: <https://dx.doi.org/10.1088/1748-9326/ac3531> (visited on 06/30/2025).
- Bureau of Transportation Statistics (2023). *National Transportation Statistics*. Tech. rep. U.S. Department of Transportation, Bureau of Transportation Statistics. URL: <https://www.bts.gov/topics/national-transportation-statistics>.
- C-4 Analytics (2025). *2025 automotive consumer price sensitivity*. URL: <https://c-4analytics.com/insights/2025-automotive-consumer-price-sensitivity>.
- Consumer Reports (2023). *Used Car Buying Guide*. URL: <https://www.consumerreports.org/cars/used-cars/buying-guide/>.
- Cox Automotive (2024). *Used vehicle retail listing prices decline in 2024*. URL: <https://www.coxautoinc.com/insights/used-vehicle-retail-listing-prices-decline-2024/>.
- (2025). *Cox Automotive car buyer journey study finds efficiency, digital tools and AI drive record satisfaction*. URL: <https://www.coxautoinc.com/insights/cox-automotive-car-buyer-journey-study-finds-efficiency-digital-tools-and-ai-drive-record-satisfaction/>.
- Fernandes, J. (2023). *US: Understanding Consumer Opinion about Used Electric Cars*. URL: <https://business.yougov.com/content/47834-us-understanding-consumer-opinion-about-used-electric-cars> (visited on 06/30/2025).
- Guzek, M., J. Jackowski, R. S. Jurecki, E. M. Szumska, P. Zdanowicz, and M. Żmuda (Jan. 2024). "Electric Vehicles—An Overview of Current Issues—Part I—Environmental Impact, Source of Energy, Recycling, and Second Life of Battery". In: *Energies* 17.1 (1), p. 249. ISSN: 1996-1073. DOI: 10.3390/en17010249. URL: <https://www.mdpi.com/1996-1073/17/1/249> (visited on 07/01/2025).
- Hagman, J., S. Ritzén, J. J. Stier, and Y. Susilo (Mar. 1, 2016). "Total Cost of Ownership and Its Potential Implications for Battery Electric Vehicle Diffusion". In: *Research in Transportation Business & Management*. Innovations in Technologies for Sustainable Transport 18, pp. 11–17. ISSN: 2210-5395. DOI: 10.1016/j.rtbm.2016.01.003. URL: <https://www.sciencedirect.com/science/article/pii/S2210539516000043> (visited on 07/01/2025).
- Helveston, J. (Oct. 20, 2025). *cbcTools: Design and Analyze Choice-Based Conjoint Experiments*. Version 0.7.1. URL: <https://cran.r-project.org/web/packages/cbcTools/index.html> (visited on 10/22/2025).
- Helveston, J. P., Y. Liu, E. M. Feit, E. Fuchs, E. Klampfl, and J. J. Michalek (2014). "Consumer Preferences for Hybrid and Electric Vehicles in China and the United States – Implications for Policy and Environment". In: *Transportation Research Board 93rd Annual Meeting*. Washington, D.C.
- Kelley Blue Book (Apr. 2025). *Average used car price rising, sales slowing*. URL: <https://www.kbb.com/car-news/average-used-car-price-rising-sales-slowing/>.
- Konstantinou, T., D. Chakraborty, and J. B. Gutierrez Lopez (2025). "Understanding the used vehicle market and its implications for electric vehicles". In: *Transportation Research Record: Journal of the Transportation Research Board* 2679.5, pp. 643–660. DOI: 10.1177/03611981241302336. URL: <https://journals.sagepub.com/doi/epub/10.1177/03611981241302336>.
- Li, L., Z. Wang, L. Chen, and Z. Wang (2020). "Consumer preferences for battery electric vehicles: a choice experimental survey in China". In: *Transportation Research Part D: Transport and Environment* 78, p. 102185. DOI: 10.1016/j.trd.2019.11.014. URL: <https://doi.org/10.1016/j.trd.2019.11.014>.
- Liao, F., E. Molin, and B. van Wee (2017). "Consumer preferences for electric vehicles: a literature review". In: *Transport Reviews* 37.3, pp. 252–275. DOI: 10.1080/01441647.2016.1230794. URL: <https://doi.org/10.1080/01441647.2016.1230794>.
- Lim, M. K., H.-Y. Mak, and Y. Rong (Feb. 2015). "Toward Mass Adoption of Electric Vehicles: Impact of the Range and Resale Anxieties". In: *Manufacturing & Service Operations Management* 17.1, pp. 101–119. ISSN: 1523-4614. DOI: 10.1287/msom.2014.0504. URL: <https://pubsonline.informs.org/doi/10.1287/msom.2014.0504> (visited on 06/30/2025).
- Loh, W. S. and R. B. Noland (2024). "Concerns expressed by used electric vehicle owners based on surveying social media". In: *Transportation Research Part D: Transport and Environment* 128, p. 104086. DOI: 10.1016/j.trd.2024.104086. URL: <https://doi.org/10.1016/j.trd.2024.104086>.
- Mashrur, S. M. and M. Mohamed (2025). "Uncovering factors affecting consumers' decisions for pre-owned electric vehicles". In: *Transportation Research Part D: Transport and Environment* 139, p. 104555. DOI: 10.1016/j.trd.2024.104555. URL: <https://doi.org/10.1016/j.trd.2024.104555>.
- Moras, B. C. K., C. M. Joslin, and K. Gkritza (2025). "Used or new electric vehicles? Public preferences and market segments". In: *International Journal of Sustainable Transportation* 20.1, pp. 129–142. DOI: 10.1080/15568318.2025.2572827. URL: <https://www.tandfonline.com/doi/full/10.1080/15568318.2025.2572827>.

- 1 Prieto, M., B. Caemmerer, and G. Baltas (2015). "Using a hedonic price model to test prospect theory assertions: the asymmetrical and nonlinear
2 effect of reliability on used car prices". In: *Journal of Retailing and Consumer Services* 22, pp. 206–212. DOI: 10.1016/j.jretconser.
3 2014.08.013.
- 4 Roberson, L. and J. P. Helveston (July 2022). "Not All Subsidies Are Equal: Measuring Preferences for Electric Vehicle Financial Incentives". In:
5 *Environmental Research Letters* 17.8, p. 084003. ISSN: 1748-9326. DOI: 10.1088/1748-9326/ac7df3. URL: [https://doi.org/10.](https://doi.org/10.1088/1748-9326/ac7df3)
6 [1088/1748-9326/ac7df3](https://doi.org/10.1088/1748-9326/ac7df3) (visited on 01/23/2026).
- 7 Rosales-Tristancho, A., A. F. Carazo, R. Brey, and H. Lindhjem (2024). "Meta-analysis of consumers' trade-offs between range and charging
8 infrastructure for electric vehicles". In: *Transportation Research Part D: Transport and Environment* 134, p. 104342. DOI: 10.1016/j.trd.
9 2024.104342. URL: <https://doi.org/10.1016/j.trd.2024.104342>.
- 10 Sheykhfard, A., M. Azmoodeh, S. Das, and B. Kutela (Jan. 1, 2025). "Analyzing Purchase Intentions of Used Electric Vehicles through Consumer
11 Experiences: A Structural Equation Modeling Approach". In: *Transport Policy* 160, pp. 125–137. ISSN: 0967-070X. DOI: 10.1016/j.
12 tranpol.2024.10.038. URL: <https://www.sciencedirect.com/science/article/pii/S0967070X24003305> (visited on
13 07/01/2025).
- 14 U.S. Environmental Protection Agency (2023). *Light-Duty Automotive Technology, Carbon Dioxide Emissions, and Fuel Economy Trends*. Tech. rep.
15 U.S. Environmental Protection Agency, Office of Transportation and Air Quality. URL: [https://cfpub.epa.gov/si/si_public_file_](https://cfpub.epa.gov/si/si_public_file_download.cfm?Lab=OTAQ&p_download_id=499454)
16 [download.cfm?Lab=OTAQ&p_download_id=499454](https://cfpub.epa.gov/si/si_public_file_download.cfm?Lab=OTAQ&p_download_id=499454).
- 17 Zou, T., M. Khaloei, and D. MacKenzie (2020). "Effects of charging infrastructure characteristics on electric vehicle preferences of new and used
18 car buyers in the United States". In: *Transportation Research Record: Journal of the Transportation Research Board* 2674.12, pp. 165–175.
19 DOI: 10.1177/0361198120952792. URL: <https://journals.sagepub.com/doi/10.1177/0361198120952792>.